

SOC Security Incident Investigation Report

Executive Summary

A simulated cyber-attack was conducted on a vulnerable test server to evaluate the organization's Security Operations Centre (SOC) detection and response capabilities. The monitoring system successfully generated an alert indicating suspicious exploitation activity. The SOC team performed timely triage, containment, and mitigation actions to prevent potential compromise.

Incident Timeline

- 11:00 AM — Suspicious exploitation activity detected
 - 11:05 AM — SOC analyst initiated alert triage
 - 11:10 AM — Source IP reputation analysis completed
 - 11:15 AM — Affected system isolated from network
 - 11:25 AM — Malicious activity blocked using security controls
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Detection and Analysis

The attack pattern indicated exploitation attempts targeting a known vulnerable network service. Security monitoring tools identified abnormal connection behaviour and generated high-priority alerts. Initial investigation confirmed unauthorized access attempts from a suspicious source IP address.

Response Actions

The SOC team immediately isolated the affected virtual machine to limit attack propagation. Network-level blocking rules were implemented to prevent further malicious communication. Endpoint security scans were performed to verify system integrity.

Recommendations

It is recommended to implement regular vulnerability patching, continuous log monitoring, and employee cybersecurity awareness programs. Deployment of advanced intrusion detection mechanisms can further enhance organizational security posture.